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Seasonal Pattern of Tomato Mosaic Tobamovirus Infection and Concentration in Red Spruce Seedlings

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Tomato mosaic tobamovirus (ToMV) infects red spruce (*Picea rubens*) and causes significant changes in its growth and physiology. The mechanism of infection and the pattern of virus concentration in seedling roots and needles were investigated. One-year-old red spruce seedlings were obtained from the nursery in April and June 1995 and August 1996 and tested for ToMV using enzyme-linked immunosorbent assay (ELISA). Virus-free seedlings were divided into three treatments: control, root inoculated, and needle inoculated. Two control, five root-inoculated, and five needle-inoculated seedlings were sampled destructively at biweekly intervals for 3 months and then tested for ToMV by ELISA. ToMV was transmitted to seedlings by root but not by needle inoculation. The virus was detected in 67 to 100% of roots but in less than 7% of needles of root-inoculated seedlings. The percent infection of root-inoculated seedlings differed significantly between the April and June and between the April and August inoculation periods. Virus concentration in infected seedling roots increased initially, peaked within 4 weeks postinoculation, and steadily declined thereafter. Significant differences in ToMV concentrations in roots also were detected among inoculation periods and sampling dates. Early spring may represent the optimal time for infection of seedlings, as well as for assaying roots for ToMV.

Tomato mosaic tobamovirus (ToMV) infects red spruce (*Picea rubens* Sarg.) across its range in the northeastern United States and impacts its growth and physiology (1, 7, 10). The virus was detected in up to 80% of dominant and codominant red spruce trees located on nine research plots in New York, Vermont, New Hampshire, and Maine (10). The concentration of ToMV in the roots of dominant-codominant red spruce on Whiteface Mountain, N.Y., was positively correlated with the number of fine roots and negatively correlated with the length of the live crown (7). Under greenhouse conditions, the infection of red spruce seedlings with ToMV was associated with (i) a 50% reduction in the rate of increase of height, weight, and root volume, (ii) reduced shoot growth, (iii) delayed budbreak, and (iv) increased freezing tolerance of current-season needles (1).

ToMV was transmitted to red spruce seedlings grown on Whiteface Mountain (10); however, the exact mechanism was not determined. Because ToMV lacks an invertebrate vector (11, 13) and seedlings did not contact native soil, Fillhart et al. (10) postulated that ToMV was transmitted by an abiotic, airborne mechanism. Infectious ToMV was detected in clouds collected from Whiteface Mountain and fog collected along the coast of Maine (6). Clouds and fog may initiate infection either by direct entry into needles through stomatal pores or by facilitating root infection through the entry of cloud (fog) condensate into soil. Direct stomatal infection of tobacco (*Nicotiana tabacum* L.) with tobacco mosaic tobamovirus (TMV) has been reported (9). In addition, both ToMV and TMV are soilborne and can infect plants, including red spruce, through their roots (1, 13, 15, 22, 25). To date, the mechanism by which airborne transmission of ToMV to red spruce occurs in natural systems remains unknown.

Little information exists concerning the replication and distribution of viruses within deciduous and coniferous forest trees. Castello et al. (5) reported variability in the detection of TMV and tobacco ringspot nepovirus in white ash (*Fraxinus americana* L.) roots and foliage sampled between May and October. The peak detection of both viruses occurred at 3- to 5-week intervals throughout the growing season. Detection of tobacco ringspot nepovirus was more consistent in roots than in foliage of white ash, but TMV was detected in equal frequency in both roots and foliage (5). Seasonal variation in the detection of poplar mosaic carlavirus (PMV) in hybrid poplar (*Populus* × *euroamericana*) also has been reported (8). The concentration of PMV in foliage generally was greater than that observed in roots or bark; however, virus distribution in symptomatic foliage was uneven. PMV was detectable in roots and phloem collected in August but not those collected in December. The virus, however, was readily detectable in terminal buds collected in December, January, and February (8).

The distribution, replication, and infection mechanism(s) of ToMV in red spruce have not yet been investigated. This knowledge will allow more accurate and precise assessment of virus infection and concentration within red spruce and perhaps other woody hosts. The objectives of this study were to determine the seasonal differences and the potential mechanisms of ToMV infection of red spruce seedlings and to examine the seasonal patterns of virus concentration within roots and needles. The four hypotheses of this study were that (i) both root and needle inoculation lead to infection of red spruce seedlings, (ii) red spruce seedlings are more susceptible to ToMV infection when inoculated in June than in April and August, (iii) ToMV concentration in roots and needles initially increases and then declines postinoculation, and (iv) mean virus concentration in roots and needles is greatest in seedlings inoculated in June.

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MATERIALS AND METHODS

Seedling inoculation. One-year-old (1-0) red spruce seedlings were obtained in April 1995 from the college nursery in Syracuse, N.Y., and tested for ToMV by double-antibody sandwich enzyme-linked immunosorbent assay (DAS-ELISA),

TABLE 1. Absorbances at 405 nm of purified ToMV dilutions and root and needle samples in three inoculation periods

Sample	Mean A_{405} ± SD for inoculation period[b]		
	April 1995	June 1995	August 1996
Purified ToMV			
100 ng/ml	1.832 ± 0.149 (18)[a]	2.000 ± 0.000 (18)	1.910 ± 0.058 (18)
10 ng/ml	0.321 ± 0.075 (18)	0.373 ± 0.024 (18)	0.303 ± 0.019 (18)
5 ng/ml	0.231 ± 0.068 (18)	0.235 ± 0.010 (18)	0.194 ± 0.011 (18)
1 ng/ml	0.155 ± 0.013 (18)	0.145 ± 0.20 (18)	0.109 ± 0.004 (18)
Negative control roots[a]	0.154 ± 0.012 (12)	0.165 ± 0.012 (12)	0.096 ± 0.009 (12)
Negative control needles[a]	0.132 ± 0.011 (24)	0.106 ± 0.004 (24)	0.091 ± 0.006 (24)
Controls			
Roots	0.167 ± 0.026 (1/12)	0.173 ± 0.021 (2/12)	0.096 ± 0.008 (0/12)
One-year-old needles	0.128 ± 0.009 (0/12)	0.105 ± 0.006 (0/12)	0.093 ± 0.006 (0/12)
Current-season needles	0.130 ± 0.006 (0/9)	0.109 ± 0.006 (0/12)	0.091 ± 0.007 (0/12)
Needle-inoculated seedlings			
Roots	0.163 ± 0.019 (2/30)	0.163 ± 0.019 (2/30)	0.096 ± 0.008 (1/30)
One-year-old needles	0.130 ± 0.012 (1/30)	0.104 ± 0.004 (0/30)	0.095 ± 0.007 (0/28)
Current-season needles	0.127 ± 0.005 (0/24)	0.108 ± 0.006 (0/30)	0.092 ± 0.008 (1/30)
Root-inoculated seedlings			
Roots	0.895 ± 0.623 (30/30)	0.414 ± 0.465 (20/30)	0.129 ± 0.041 (21/30)
One-year-old needles	0.130 ± 0.012 (1/30)	0.104 ± 0.005 (0/30)	0.098 ± 0.018 (2/29)
Current-season needles	0.128 ± 0.006 (0/25)	0.108 ± 0.005 (1/30)	0.092 ± 0.007 (1/30)

[a] Roots or needles from 1-year-old, virus-free red spruce seedlings were collected from the college nursery and triturated 1/5 in ELISA extraction buffer.
[b] Single values in parentheses are number of wells. Other values in parentheses are numbers of samples positive/total number. Samples were considered positive if the mean of replicate sample wells was greater than the mean of absorbance of control root or needle extracts plus three times the standard deviation.

as described by Bachand et al. (1). A group of 125 seedlings that tested negative for ToMV were selected and subdivided into three treatments: control, root-inoculated, and needle-inoculated seedlings (n = 25, 50, and 50, respectively).

Control and needle-inoculated seedlings were transplanted into 120-mm-diameter pots containing a sterilized mixture of Promix (Premier Horticulture, Inc., Rivière-du-Loup, Quebec, Canada) and sand (3:1 [vol/vol]). Plastic trays were placed under each pot to prevent root contact with both the surface of greenhouse benches and the roots of neighboring seedlings. Seedlings were watered daily with autoclaved, deionized water for the duration of the experiment. Needle inoculation was performed by misting 50 seedlings weekly for 3 months using a plastic, hand-held garden sprayer (Sprayco, Detroit, Mich.) containing a 50-ng/ml solution of ToMV-38, a stream water isolate from Whiteface Mountain (14), prepared in autoclaved, deionized water. The inoculum was adjusted to a pH of 4.0 to 5.0 with dilute sulfuric acid. A plastic bag was placed around each pot and on the soil surface during needle inoculation to prevent deposition of the mist or condensate on the pot or soil surface. Seedlings were misted until visible droplets formed on the needles. Inoculum infectivity was confirmed by mechanical inoculation of *Chenopodium quinoa* Willd. leaves with each preparation. Fifty root-inoculated seedlings were grown for 1 week in a hydroponic culture of dilute nutrient solution (28) containing ToMV-38 (50 ng/ml) and 0.025% Benomyl. The pH of the solution was adjusted to approximately 4.0 with dilute sulfuric acid and then monitored daily. The infectivity of the ToMV-containing nutrient solution also was confirmed by mechanical inoculation of *C. quinoa* leaves. Following the 1-week incubation period, root-inoculated seedlings were removed from hydroponic culture, thoroughly rinsed with tap water, and transplanted as described above. Control seedlings were neither root nor needle inoculated. All seedlings were maintained in a greenhouse under ambient temperature and day length conditions and were fertilized weekly with dilute nutrient solution.

The above procedures were repeated using seedlings that were obtained from the nursery in June and August 1995. Therefore, three inoculation periods were utilized for this experiment: April, June, and August. A high rate of mortality due to transplantation-associated shock occurred in August 1995; thus, the August inoculation experiment was repeated in 1996.

Sampling procedure and DAS-ELISA. Two control, five root-inoculated, and five needle-inoculated seedlings were randomly selected at biweekly intervals for 12 weeks within each inoculation period and then destructively sampled. Roots, current-season needles, and 1-year-old needles were removed, rinsed thoroughly with tap water, placed into separate 1.5-ml microcentrifuge tubes, and stored at −20°C. Approximately 30 to 100 mg of roots and needles were triturated in liquid nitrogen, diluted 1/5 in ELISA extraction buffer, and tested by DAS-ELISA, as previously described (1).

Data analyses. Root and needle samples were considered positive for ToMV if the mean absorbance of replicate wells was greater than the mean plus three times the standard deviation of four wells containing virus-free root or needle extracts. Fisher's exact test was used to determine if (i) root and/or needle inoculation resulted in infection of seedlings in each inoculation period and (ii) the frequency of seedling infection varied among inoculation periods (26).

To determine the virus concentration of each sample, a standard curve was prepared by using multiple regression analysis, a quadratic polynomial model, and absorbances of purified ToMV dilutions (100, 10, 5, and 1 ng/ml) and virus-free root and needle extracts. Concentrations then were multiplied by 5 to correct for tissue dilution. Mean virus concentrations (nanograms of virus per gram of tissue) per seedling were calculated using all seedlings, positive and negative, within each treatment. Treatment means were plotted as a function of sampling date.

Univariate analysis of variance (ANOVA) was used to determine if the ToMV concentration in roots and/or needles differed among the three treatments (control, needle inoculation, and root inoculation). Sampling time was treated as a blocking factor in these analyses. Further, each inoculation period (i.e., April, June, and August) was analyzed separately.

Univariate split-plot ANOVA was used to determine if ToMV concentration differed among inoculation periods and sampling times (21, 26). Because ToMV was detected consistently only in the root samples of root-inoculated seedlings, the split-plot analysis was performed with only data from these seedlings. Inoculation period and sampling time were treated as the whole-plot and subplot factors, respectively. Orthogonal contrasts were used to compare the linear and quadratic components of the inoculation periods. Linear and "lack of fit" contrasts were used to examine the linear and nonlinear components of the interaction term (26).

RESULTS

Transmission of ToMV to seedlings. ToMV was transmitted to red spruce seedlings by root inoculation but not by needle inoculation in all three inoculation periods (Tables 1 and 2). The sensitivity of the ELISA was between 5 and 25 ng/g (1 and 5 ng/ml) (Table 1). All *C. quinoa* plants inoculated with ToMV-containing nutrient solution and inoculum for needle inoculation displayed necrotic local lesions, typical of ToMV, at 5 to 7 days postinoculation. The virus was detected in the

TABLE 2. *P* values calculated with Fisher's exact test for comparison of percent ToMV detection in red spruce roots and needles among three treatments in each of three inoculation periods

Inoculation period and sample	*P* value for detection in:		
	Control vs. root-inoculated samples	Control vs. needle-inoculated samples	Root-inoculated vs. needle-inoculated samples
April 1995			
Roots	<0.001	0.455	<0.001
One-year-old needles	0.714	0.714	0.508
Current-season needles	1.000	1.000	1.000
June 1995			
Roots	<0.001	0.256	<0.001
One-year-old needles	1.000	1.000	1.000
Current-season needles	0.714	1.000	0.500
August 1996			
Roots	<0.001	0.714	<0.001
One-year-old needles	0.485	1.000	0.237
Current-season needles	0.714	0.714	0.508

roots of 8, 6, and 79% of control, needle-inoculated, and root-inoculated seedlings, respectively, across all three inoculation periods combined. ToMV was detected in the needles (current-season and 1-year-old needles) of 0, 1, and 3% of control, needle-inoculated, and root-inoculated seedlings, respectively, across all three inoculation periods combined. The levels of detection of ToMV in roots differed significantly among the three treatments in each inoculation period ($P < 0.001$) (Table 2). The virus was detected more frequently in the roots of April root-inoculated seedlings than in those root inoculated in either June or August ($P < 0.001$) (Table 1). However, the frequency of detection did not differ between June and August root-inoculated seedlings ($P = 0.210$) (Table 1). The levels of detection of ToMV in current-season and 1-year-old needles did not differ significantly among the three treatments ($P > 0.230$) or inoculation periods ($P > 0.210$) (Table 2).

TABLE 3. Summary of univariate ANOVA comparing ToMV concentrations in the roots of control, root-inoculated, and needle-inoculated seedlings (treatments) inoculated in April, June, and August

Month and source	ANOVA data			
	df	MS[b]	*F*	*P*
April				
Treatment	2	388,280.2	13.629	0.001
Block[a]	5	26,993.0	0.947	0.492
Error	10	28,489.3		
Total	17			
June				
Treatment	2	39,926.5	4.994	0.031
Block	5	10,067.3	1.259	0.353
Error	10	7,995.6		
Total	17			
August				
Treatment	2	232.9	4.506	0.040
Block	5	27.014	0.523	0.755
Error	10	51.699		
Total	17			

[a] Sampling time was treated as a blocking factor for these analyses.
[b] MS, mean square.

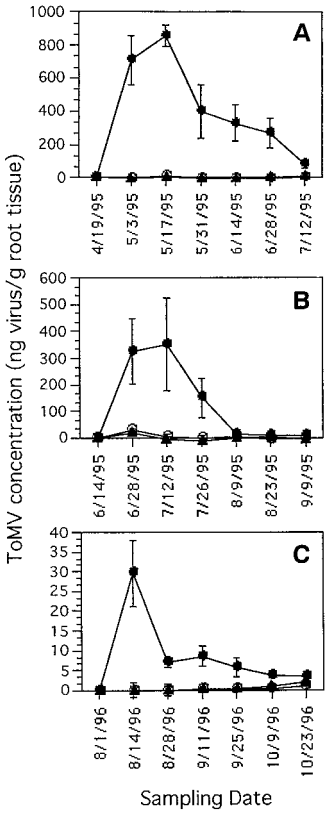


FIG. 1. Mean ToMV concentration in the roots of control (○), needle-inoculated (▲), and root-inoculated (●) seedlings at six biweekly sampling times in each of three inoculation periods: April (A), June (B), and August (C). Bars represent 1 standard error of the mean.

Differences in ToMV concentrations among treatments. The mean virus concentrations in the roots of control, needle-inoculated, and root-inoculated seedlings were 1.2, 0.8, and 440.8 ng/g, respectively, for the April inoculation period. The mean virus concentrations in the roots of control, needle-inoculated, and root-inoculated seedlings were 4.2, 0.3, and 143.2 ng/g, respectively, for the June inoculation period. The mean virus concentrations in the roots of control, needle-inoculated, and root-inoculated seedlings were 0.1, 0.1, and 10.8 ng/g, respectively, for the August inoculation period. A significant difference in the virus concentrations in roots was detected among the three treatments (Table 3; Fig. 1), but this was not found for the current-season or 1-year-old needles (data not shown) for all three inoculation periods.

Variation in ToMV concentration over time and among inoculation periods. A significant difference in ToMV concentration in roots of root-inoculated seedlings was detected among the three inoculation periods (Tables 4 and 5; Fig. 2). The mean ToMV concentrations in roots across all six sampling dates of seedlings root inoculated in April, June, and August were 440.8, 143.2, and 10.8 ng/g, respectively. The virus concentration in the roots of root-inoculated seedlings fluctuated over the 12-week sampling period in each of the three

TABLE 4. Summary of split-plot ANOVA for the comparison of ToMV concentrations in the roots of seedlings that were root inoculated in April, June, or August

Source	df	MS[a]	*F*	*P*
Time of inoculation	2	803,154.7	27.214	<0.001
Error A	12	29,512.66		
Sampling time	5	588,656.6	16.767	<0.001
Time of inoculation × sampling time	10	111,349.0	3.172	0.003
Error B	60	35,108.92		
Total	89			

[a] MS, mean square.

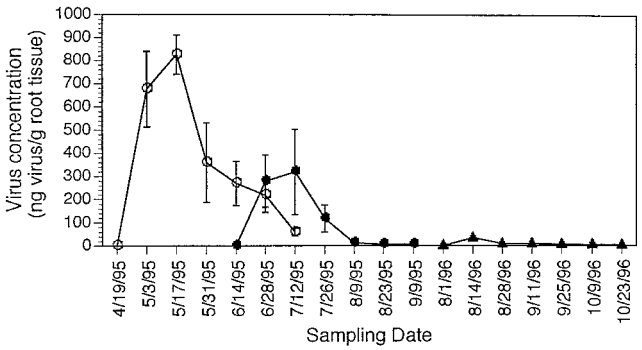


FIG. 2. Mean ToMV concentration in roots of root-inoculated seedlings at six biweekly sampling times in each of three inoculation periods: April (○), June (●), and August (▲). Bars represent 1 standard error of the mean.

inoculation periods. In both the April and June inoculation periods, ToMV concentration in roots increased initially, peaked at 4 weeks postinoculation, and then steadily declined over time (Fig. 1A and B). In the August inoculation, the ToMV concentration in the roots peaked 2 weeks postinoculation, then declined, and finally remained steady at approximately 10 ng/g (Fig. 1C). A significant difference in virus concentration in roots of root-inoculated seedlings was detected among sampling times ($P < 0.001$) (Table 4). Significant variation in the linear ($P = 0.002$), but not the quadratic ($P = 0.624$), component of the sampling time factor was detected (Table 5). A significant interaction between inoculation period and sampling time also was detected in roots of root-inoculated seedlings ($P = 0.003$) (Table 4). Further, the linear portion of the virus concentration curves differed among the inoculation periods ($P < 0.001$) (Table 5).

DISCUSSION

Needle inoculation of red spruce seedlings did not result in ToMV infection (Tables 1 and 2). Fillhart et al. (10) reported the airborne transmission of ToMV to red spruce seedlings on Whiteface Mountain, and postulated that transmission was abiotic and occurred through contact with virus-laden clouds. The physiognomy of red spruce allows for the efficient interception of clouds and/or fog and therefore of ToMV. Further, high rates of fog deposition on red spruce needles have been reported, particularly on the epicuticular waxes that occlude epistomatal chambers (17). The entry of cloud and fog deposition, and possibly of ToMV, into stomates through mass flow is possible; however, infection of surrounding mesophyll cells and entry into the symplast are more difficult barriers. Although the transmission of TMV to tobacco was demonstrated by a similar method (9), inherent differences between conifer needles and tobacco leaves may account for the lack of infection observed in our experiment. Alternatively, conditions utilized in our experiment do not precisely simulate those occurring in nature. Factors such as droplet size, inoculum concentration, impact velocity, wounding, and pH may affect the transmission of ToMV to seedlings by needle inoculation. Virus-laden cloud and/or fog deposition also may condense and fall to the soil surface, as opposed to entering stomatal chambers, and so facilitate root infection of red spruce. This mode of transmission, however, was prevented in our experiment by covering the soil surface with plastic.

TABLE 5. Summary of main effect and interaction contrasts for the comparison of ToMV concentrations in the roots of seedlings that were root inoculated in April, June, or August

Contrast	Contrast df	Error df	*F*	*P*
Sampling time				
Linear	1	60	10.214	0.002
Quadratic	1	60	0.243	0.624
Interaction				
Linear	2	60	13.548	<0.001
Lack of fit	8	60	0.577	0.792

ToMV was transmitted to red spruce seedlings by root inoculation with hydroponic culture (Tables 1 and 2). In prior experiments, ToMV was transmitted to 35 to 85% of red spruce seedlings that were incubated for 24 h in a 1-μg/ml solution of purified ToMV and Celite (1, 15). Celite was utilized to induce wounds and provide infection courts. In this study, ToMV was detected in 67 to 100% of seedlings that were root inoculated with a nutrient solution containing 50 ng of purified ToMV per ml without abrasive additives. Wounding due to transplantation and handling, however, may have facilitated infection. Tomato (*Lycopersicon esculentum* L.) and pepper (*Capsicum annuum* L.) grown in ToMV-containing hydroponic culture become infected readily (22, 25). Efficient transmission of maize white line mosaic virus to maize (*Zea mays* L.) seedlings also has been achieved with hydroponic culture (19). Our study represents the first report of virus transmission to a forest tree species in a hydroponic system. The transmission of viruses to forest tree species, especially conifers, is difficult. Our hydroponic procedure provided an easy and efficient method for virus transmission to red spruce seedlings.

ToMV was detected frequently (>67%) in the roots but infrequently (<7%) in the needles of infected red spruce seedlings (Table 1). The concentration of ToMV in roots also was considerably greater than in needles (Table 1). The virus was purified and transmitted from the needles of mature, dominant and codominant red spruce trees on Whiteface Mountain, but the frequency and concentration of ToMV were much lower than in the roots (16). Thus, replication of ToMV in red spruce occurs primarily in root tissues and does not occur or is limited in needle tissues. The presence of virus in needles may represent either a low level of virus replication or passive movement of virus with plant assimilates, nutrients, and water from roots to needles. A more detailed investigation (e.g., detection of the viral replicative intermediate) is required to determine the exact location of the replicating ToMV in spruce tissue. At the tissue level, ToMV was detected in cortical and epidermal cells, as well as in the lateral root primordia of root-inoculated seedlings, by immunofluorescence (2). In recent experiments, both the positive and negative strands of ToMV have been

detected in root-inoculated red spruce and white pine seedlings by reverse transcription-PCR (4a).

The time of inoculation affects both the transmission and the concentration of ToMV in the roots of red spruce seedlings (Table 4 and Fig. 2). To our knowledge, our study represents the first report of the seasonal effects on virus infection and concentration in the roots of any woody plant species. Accurate and precise assessment of infection is dependent upon virus concentration. The sensitivity of our ELISA system was 5 to 25 ng/g of root or needle tissue (Table 1). Therefore, roots and needles in which the virus concentration did not exceed 5 to 25 ng/g would be assessed incorrectly as negative. The virus concentration in the roots of infected seedlings was considerably lower in seedlings inoculated in August than in those inoculated in either April or June (Fig. 2). The concentration of ToMV observed in seedlings inoculated in August rarely exceeded 10 to 25 ng/g (Fig. 1). Thus, infected seedlings may have been assessed as negative and so may account for the observed differences in percent infection of root-inoculated seedlings among the inoculation periods. A more sensitive assay, e.g., reverse transcription-PCR, may be required to further investigate seasonal effects on virus transmission.

The virus concentration in the roots of ToMV-infected seedlings peaked within 4 weeks postinoculation and subsequently decreased over time at a linear rate (Table 5; Fig. 2). The rate of decrease in virus concentration (i.e., the linear portion of the concentration curve) differed significantly among the three inoculation periods (Table 5). A similar pattern in virus concentration was observed in pepper foliage systemically infected with cucumber mosaic cucumovirus. The concentration of cucumber mosaic cucumovirus in symptomatic foliage peaked approximately 1 week postinoculation and steadily decreased with time (18). This pattern also has been observed in soybean [*Glycine max* (L.) Merr.] foliage inoculated with soybean dwarf luteovirus (12).

Seasonal effects on virus concentration and disease development have been reported in a number of herbaceous and woody hosts (5, 23–25, 29). The mean ToMV concentration in roots was greatest in seedlings root inoculated in April, followed by those root inoculated in June and August (441, 143, and 11 ng/g, respectively) (Fig. 2). *Prunus* necrotic ringspot, prune dwarf, and apple mosaic ilarviruses were detected by ELISA most frequently in the infected foliage of apple (*Malus domestica* Borkh.), plum (*Prunus domestica* L.), and cherry (*Prunus* spp.) trees early in the growing season (29), suggesting a decrease in concentration of these viruses during the growing season. Similarly, *Prunus* necrotic ringspot and apple mosaic ilarviruses were detected easily in infected almond [*Prunus dulcis* (Mill.) Webb] early in the season but could not be detected after June (4). The relative concentration of plum pox potyvirus in peach [*Prunus persica* (L.) Batsch] leaves also peaked early in the season (i.e., May), and then steadily declined (23). Seasonal effects on the concentration of grapevine fanleaf nepovirus in infected plants also have been observed (24). A high concentration of grapevine fanleaf nepovirus was detected in mature leaves in May, followed by a rapid decrease to a low, constant level for the remainder of the growing season (24). Seasonal differences in the root necrosis and wilting of pepper (*C. annuum* cv. Hungarian Wax) infected with ToMV have been observed in hydroponic cultures (25). The observed differences in disease severity of ToMV-infected pepper were associated with both temperature and relative humidity (25). Day length and light intensity also may affect disease expression and virus replication (20).

Physiological age and developmental stage of a host are important factors that influence the course of infection and disease development (20). Age-dependent susceptibility of *N. tabacum* cv. Samsun and Samsun NN to TMV infection has been reported (27). Reduced susceptibility of soybean to soybean dwarf luteovirus infection was correlated with plant age (10). ToMV infection and concentration in roots were greatest in seedlings inoculated in April, when the roots were dormant. Reduced competition for cellular constituents and metabolites in dormant tissues may allow the virus to reach high concentrations. Ribosome content was associated with mature plant resistance in potato against potato X potexvirus (30); however, a causal relationship was not established. Endogenous host compounds such as hormones also may influence virus replication (19). For example, Balázs et al. (3) reported increased TMV multiplication in tobacco (*N. tabacum* cv. Xanthi-nc) leaves treated with abscisic acid.

Understanding the seasonal patterns of ToMV infection and replication is important for evaluating the ecological aspects of this pathosystem and permits a more accurate and precise assessment of virus infection and concentration in this and perhaps other woody hosts. Infections of seedlings and ToMV concentration in roots were greatest when root inoculation was performed prior to the onset of root growth. Early spring may represent an important window for ToMV infection of red spruce in natural ecosystems, as well as the ideal time to assay tissue for virus infection. Additional investigation, however, is required to determine the specific seasonal and developmental factors (e.g., day length, temperature, humidity, and physiological age) that affect virus infection and replication.

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